Parth Dhengle

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Summary

Final-year B.Tech (AI & DS) student with hands-on experience building **AI systems, full-stack applications, and data pipelines**. Published researcher in AI-generated image watermarking. Currently deepening expertise in **MLOps and Data Engineering**. Open to roles in **AI/ML Engineering or Data Engineering**.

Education

AISSMS Institute of Information Technology, Pune — **Aug 2023 – May 2027**

B.Tech in Artificial Intelligence & Data Science | ***CGPA: 8.62 / 10.0***

Experience

YCS Tech | SDE Intern — **Aug 2025 – May 2026**

Pune, Maharashtra | Full-time, On-site

- **Spearheaded end-to-end development** of an R&D Battery Simulation Tool, taking full ownership of the system from prototype to final client delivery.
- Migrated **MATLAB simulation code to Python** (NumPy, Pandas) and secured it with a **pytest** suite, reducing runtime from 30 to 10 mins (**~67% faster**).
- Architected the stack (**FastAPI**, **Next.js**) with an optimized simulation loop, caching, and packaged it as a **Windows desktop app** via Electron.
- Replaced MongoDB with a **JSON file-based storage** layer, strategically prioritizing core simulation performance over raw data fetch speeds.

Projects

SmartShelf – MLOps Platform | *XGBoost, LightGBM, FastAPI, Prefect, MLflow, PostgreSQL, BigQuery, Docker*

- Designed a **chained 3-model pipeline** where each model's prediction becomes a feature for the next – Demand Forecast → Price Optimisation → Inventory EOQ built on a normalised **PostgreSQL** OLTP schema I architected from scratch.
- Set up **BigQuery** as an OLAP layer for analytical queries, automated weekly retraining via **Prefect**, and tracked all experiments and model versions in **MLflow Registry**.
- Served predictions via **FastAPI**, added **Prometheus + Grafana** monitoring, and containerised the full system with **Docker** and **GitHub Actions** CI/CD.

NOVA – Agentic AI System | *FastAPI, CrewAI, FAISS, Electron, Firebase*

- Developed an **autonomous multi-agent system** that translates high-level user goals into multi-step, executable Python workflows without manual intervention.
- Implemented a **context-aware RAG pipeline** (FAISS) for document indexing, packaging the system as an **Electron** desktop app with Firebase authentication.

GradeX – AI Coding Platform | *Next.js, Node.js, AWS EC2, OpenRouter, Docker*

- Built a **1v1 competitive coding platform** featuring **LLM-generated** problems via OpenRouter and sandboxed code execution on **Docker and AWS EC2**.
- Implemented **ELO matchmaking** algorithm from scratch to rank players and balance duel pairings in real time.

Research & Publications

Invisible Watermarking for AI-Generated Images – IJSREM — **April 2026**

Co-author: Atharva Deo | Guide: Dr. D.S. Zingade, AISSMS IOIT

- First comparative study of **Spatial vs. Transform Domain** watermarking on **Stable Diffusion XL** outputs for deepfake detection, IP protection, and AI content authenticity.

Achievements

- **3× Winner** – Coding Relay (AISSMS, Keystone) and Blind Coding at inter-college fests.
- **Top-3 Finishes** – 2nd in Hack the Template & Code Hunt; 3rd in Bit by Bit. **7+ hackathons** participated.

Technical Skills

Languages: Python, C++, JavaScript, TypeScript, SQL
AI / ML: Machine Learning, RAG, XGBoost, LightGBM, scikit-learn, FAISS
MLOps & Data: MLflow, Prefect, ETL Pipelines, Feature Engineering, Evidently, Prometheus, Grafana, BigQuery
Frameworks & DBs: FastAPI, Next.js, React, CrewAI, PostgreSQL, MongoDB, Firebase
DevOps & Tools: Docker, CI/CD (GitHub Actions), Git, REST APIs, AWS EC2, Electron
Core CS: DSA, System Design, OOP, OS, Computer Networks

Leadership & Responsibility

- **ML Head, AI & DS Dept.** – Organised ML/GenAI workshops and mentored peers on AI projects.
- **Published Researcher** – Co-authored a peer-reviewed paper on AI content authentication and digital watermarking.